

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1482

North Bonneville 100 kV
July 31, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1482. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the North Bonneville 100 kV substation in the Illinois area. The request was received on May 26, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 351 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$99,570,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
NORTH BONNEVILLE – LAKE CHARLES PLANT 100kV	OXBOW – N BONNEVILLE 100kV	1326.2	114	100.0
CLINTMOORE – LAKE CHARLES PLANT 100kV Line	OXBOW – N BONNEVILLE 100kV	802.3	179	100.0
CEDAR – DECATUR ROUTE 51 100kV	OXBOW – N BONNEVILLE 100kV L	302.3	136	19.6
DECATUR ROUTE 51 – ASBURY 100kV	OXBOW – N BONNEVILLE 100kV	299.6	134	19.6
BRANDY BRANCH – ASBURY 100kV	OXBOW SUBSTATION – NORTH BONNEVILLE 100kV	298.5	131	19.6
LEOTA SUBSTATION – BRANDY BRANCH 100kV line	OXBOW – N BONNEVILLE 100kV L	296.3	128	19.6
FLAMBEAU HYDROELECTRIC STA 100-1 kV Auto	OXBOW – N BONNEVILLE 100kV	267.2	240	47.3
FLAMBEAU HYDROELECTRIC STA – CEDAR 100kV	NORTH BONNEVILLE – OXBOW 100kV L	253.1	170	19.6
SHRELINE – FLAMBEAU HYDRELECTRIC STA 100kV	OXBOW – N BONNEVILLE 100kV	249.3	520	81.2
SHOREINE – FRONT RANGE 100kV	OXBOW – N BONNEVILLE 100kV	244.0	539	81.2
SOCRATES 19 – FRONT RANGE 100kV Line	OXBOW – NORTH BONNEVILLE SUBSTATION 100kV line	244.0	539	81.2
FLAMBEAU HYDROELECTRIC STA – DAVIS CREEK 100kV	OXBOW SUBSTATION – NORTH BONNEVILLE 100kV	240.1	103	14.3
ANZA 230/100 kV Transformer Bank	FRONT RANGE – SHORELINE 100kV	232.8	135	15.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BEAR CANYON – SOCRATES 19 100kV Line	OXBOW SUBSTATION – N BONNEVILLE 100kV	226.4	667	99.9
OXBOW – BLYTHE 100kV line	CLINTMOORE – BEAR CANYON 100kV	225.2	673	99.9
BEAR CANYON – CLINTMOORE 100kV line	OXBOW – NORTH BONNEVILLE 100kV	222.9	686	99.9
OXBOW – N BONNEVILLE 100kV Line	BEAR CANYON – CLINTMOORE 100kV	222.9	686	99.9
FLAMBEAU HYDROELECTRIC STA 230/1kV Transformer	OXBOW – N BONNEVILLE 100kV	214.7	290	47.3
TOCOI – SHAWBORO 100kV line	SHORELINE – FRONT RANGE 100kV	198.6	207	27.2
OREGON CITY – TOCOI 100kV	SHORELINE – FRONT RANGE 100kV	196.8	211	27.2
OREGON CITY – KENZIE CREEK 100kV	FRONT RANGE – SHORELINE 100kV	194.8	215	27.2
KENZIE CRK – SOCRATES 19 SUBSTATION 100kV	FRONT RANGE SUBSTATION – SHORELINE 100kV	185.0	240	27.2
SWEATT – TINLEY PARK 100kV Line	LAWRENCE HILL – SEATT 100kV Line	179.6	132	9.3
PRICE HILL – DAVIS CREEK 100kV L	OXBOW SUBSTATION – N BONNEVILLE SUBSTATION 100kV	170.9	125	14.3
SHAWBORO – SKY VALLEY 100kV	SHORELINE SUBSTATION – FRONT RANGE 100kV	170.0	119	13.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NORTH BONNEVILLE – LAKE CHARLES PLANT 100kV	Rebuild line with higher-capacity conductor	\$5,952,000	100% (\$5,952,000)
LAKE CHARLES PLANT – CLINTMOORE 100kV line	Rebuild line with higher-capacity conductor	\$7,157,000	100% (\$7,157,000)
CEDAR – DECATUR ROUTE 51 100kV	Rebuild line with higher-capacity conductor	\$1,797,000	88% (\$1,578,000)
ASBURY – DECATUR ROUTE 51 100kV	Rebuild line with higher-capacity conductor	\$2,559,000	88% (\$2,247,000)
BRANDY BRANCH – ASBURY 100kV	Rebuild line with higher-capacity conductor	\$2,417,000	88% (\$2,122,000)
BRANDY BRANCH – LEOTA 100kV	Rebuild line with higher-capacity conductor	\$6,246,000	88% (\$5,485,000)
FLAMBEAU HYDROELECTRIC STATION 100/1 kV Autotransformer	Replace transformer with higher-rated unit	\$3,500,000	86% (\$3,024,000)
FLAMBEAU HYDROELECTRIC STA – CEDAR 100kV	Rebuild line with higher-capacity conductor	\$412,000	88% (\$361,000)
SHORLINE – FLAMBEAU HYDROELECTRIC STATION 100kV	Rebuild line with higher-capacity conductor	\$669,000	90% (\$599,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SHOREINE – FRONT RANGE 100kV	Rebuild line with higher-capacity conductor	\$1,638,000	90% (\$1,466,000)
SOCRATES 19 – FRONT RANGE 100kV Line	Rebuild line with higher-capacity conductor	\$1,684,000	90% (\$1,507,000)
FLAMBEAU HYDROELECTRIC STATION – DAVIS CRK 100kV	Rebuild line with higher-capacity conductor	\$1,251,000	92% (\$1,150,000)
ANZA 230-100 kV TX	Replace transformer with higher-rated unit	\$5,726,000	67% (\$3,822,000)
SOCRATES 19 – BEAR CANYON 100kV	Rebuild line with higher-capacity conductor	\$4,023,000	97% (\$3,906,000)
OXBOW – BLYTHE 100kV line	Rebuild line with higher-capacity conductor	\$1,898,000	97% (\$1,842,000)
BEAR CANYON – CLINTMOORE 100kV line	Rebuild line with higher-capacity conductor	\$5,254,000	97% (\$5,101,000)
OXBOW – N BONNEVILLE 100kV	Rebuild line with higher-capacity conductor	\$6,964,000	97% (\$6,761,000)
FLAMBEAU HYDROELECTRIC STA 230/1kV Transformer	Replace transformer with higher-rated unit	\$12,532,000	86% (\$10,828,000)
TOCOI – SHAWBORO 100kV line	Rebuild line with higher-capacity conductor	\$5,813,000	83% (\$4,821,000)
OREGON CITY – TOCOI 100kV	Rebuild line with higher-capacity conductor	\$5,210,000	83% (\$4,321,000)
OREGON CITY – KENZIE CRK 100kV	Rebuild line with higher-capacity conductor	\$10,276,000	83% (\$8,522,000)
KENZIE CRK – SOCRATES 19 SUBSTATION 100kV	Rebuild line with higher-capacity conductor	\$2,071,000	83% (\$1,717,000)
TINLEY PARK – SWEATT 100kV	Rebuild line with higher-capacity conductor	\$5,791,000	98% (\$5,674,000)
PRIC HILL – DAVIS CREEK 100kV	Rebuild line with higher-capacity conductor	\$2,874,000	92% (\$2,642,000)
SHAWBORO – SKY VALLEY 100kV	Rebuild line with higher-capacity conductor	\$8,590,000	81% (\$6,965,000)
	Total allocated to project		\$99,570,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.